Predicting Heart Disease Using Machine Learning Techniques (English Translation)

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Tartous University

Faculty of Information and Communication Technology Engineering

Department of Computer and Electronic Systems

[University logo]

A graduation project prepared for the Bachelor's degree in Information and Communication Technology Engineering

Predicting Heart Diseases Using Machine Learning Techniques

Predicting Heart Disease Using Machine Learning Techniques

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Page 2 — Dedication

[Dedication page (إهداء). The text of this page is garbled in the PDF's text layer and could not be read reliably, so it is not translated here.]

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Purpose of the Project

The project aims to build an integrated intelligent system for reading and diagnosing heart diseases using advanced machine learning techniques. The project includes developing medical data analysis algorithms and deep learning for analyzing medical images such as the electrocardiogram (ECG) and radiological images. The system aims to provide an accurate and rapid diagnosis for a variety of heart diseases, helping physicians make more informed treatment decisions. The project also seeks to improve early detection rates of heart diseases and reduce the rate of medical errors by providing recommendations based on big data and advanced analysis models. In addition, the system aspires to provide a user-friendly interface for physicians and health practitioners, facilitating the adoption of the technology in various medical environments.

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Introduction

Cardiovascular diseases have long been among the leading causes of death worldwide. Among these diseases, myocardial infarction (MI), known as a heart attack, is one of the most serious conditions, as it causes the heart muscle to fail to contract sufficiently for a long period. Rapid therapeutic intervention within the first hour of a heart attack can significantly reduce the chances of death. When physicians face an emergency heart condition, the electrocardiogram (ECG) is considered the first and most important diagnostic tool used. The ECG monitors the heart's electrical activity and converts it into a graph reflecting the periodic electrophysiological changes in the heart muscle, which helps identify any possible disturbances. Accurate analysis of the electrocardiogram can enable physicians to diagnose myocardial infarction. However, manual diagnosis faces challenges related to sensitivity and accuracy, as its sensitivity is 91% and its specificity only 51%. Hence comes the importance of developing a computer-aided system for the automatic detection of MI, which can enhance the accuracy and speed of diagnosis, helping cardiologists make better treatment decisions. In recent years, given the complexity of classifying heart disorders, neural network techniques have been increasingly adopted in research and applications for the automatic detection of myocardial infarction.

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Chapter One

(The Human Heart Anatomically and Electrically)

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1-1 Anatomy of the Heart

The human heart is the essence of human life, and from here stems the importance of studying the cardiac system as a whole, in terms of its physiological and anatomical structure, its mechanism of action, the diseases that affect it, and how to detect them.

The heart is a hollow muscular organ about the size of a fist. It is located in the middle of the chest, protected by the bones of the rib cage at the front and by the spine at the back. It is cradled by the lungs and rests on the diaphragm, which separates it from the abdominal organs, and its lower tip is inclined toward the left side.

[Figure: chest and heart position. Labels: Second intercostal space; Fifth intercostal space; Apex; Midline]

Physiologically, the heart is described as an involuntary striated muscle that acts as a pump, ensuring that blood is pumped to the arteries spread throughout the body and thus meeting the needs of all cells in the body for the essential elements required for their survival, such as food and oxygen. On the other hand, the cardiac pump pumps blood loaded with carbon dioxide to the lungs, where it is freed of this gas and loaded with pure oxygen. The heart is divided into two separate halves, namely the right and left sides of the heart.

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- **The right side:** pumps blood through the lungs.
- **The left side:** pumps blood through the surrounding organs.

A muscular wall that prevents the blood from mixing between these two parts separates them, and it is called the septum.

[Figure: Internal View of the Heart. Labels: Pulmonary valve; Aorta (arch); Pulmonary arteries; Pulmonary veins; Left atrium; Right atrium; Mitral valve; Tricuspid valve; Aortic valve; Right ventricle; Left ventricle]

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1-2 The Cardiac Cycle

It is the period extending from the start of one heartbeat to the start of the next, and the cardiac cycle passes through the following phases.

[Figure: Pulmonary circulation / Systemic circulation. Labels: Pulmonary Artery; Pulmonary Veins; Venae Cavae; Aorta; RA; LA; RV; LV]

- **Atrial contraction phase:**

Atrial contraction begins near the opening of the superior vena cava and spreads rapidly to the right atrium and then the left. It takes a period of (0.1 s) when the heart rate is (72 PULSE/MIN), and in this phase the P wave is drawn.

- **Ventricular contraction phase:**

In this phase the atrioventricular valves close because of the sudden rise in ventricular pressure, the aortic and pulmonary valves open, the waves (Q, R, S) are drawn, and the beginning of the T wave up to its peak, lasting about (0.35 s).

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- **Relaxation phase:**

The aortic and pulmonary valves close and the atrioventricular valves open. In this phase the remaining part of the T wave and the U wave are drawn, and this takes (0.4 s). Thus we note that the cardiac cycle takes about (0.8 s) and that the heart works half of the time and relaxes half of the time, which makes the heart pump blood continuously.

1-3 The Electrical System of the Heart

The heart muscle has automaticity: it is able to generate the impulses needed for its work by itself, and it also possesses a specialized system able to transmit these impulses very quickly.

The conducting electrical system consists of:

- The internodal pathways.
- The sinus node.
- The atrioventricular node.
- The atrioventricular bundle.
- The left bundle branch.
- The right bundle branch.

It was long believed that atrial activation occurs by a successive advance of the stimulation wave from the sinoatrial node toward the periphery, but studies have shown that there are special pathways through which the stimulation wave is conducted, while the other tissues show high electrical resistance. Located in the upper lateral wall of the right atrium, below the opening of the superior vena cava, is the atrial lateral node, which is called the Keith and Flack node.

The atrioventricular node passes stimuli in one direction only, from the atria to the ventricles. The bundle of His arises from the atrioventricular node and crosses the fibrous skeleton of the heart, forming the only pathway for conducting the stimulus between the atria and the ventricles. After that it divides into two branches, right and left, and then the left branch divides into two parts, anterior and posterior. These two branches run through the septum between the ventricles, directly beneath the endocardium, until they reach the apex of the heart. Purkinje fibers branch from the two branches to transmit the stimulus to the muscle fibers, and thus the stimulus always begins from the endocardium toward the pericardium.

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[Figure: conduction system. Labels: Sinoatrial node; Bachmann′s bundle; Atrioventricular node; His bundle; Left posterior bundle; Right bundle; Purkinje fibers]

1-4 Principle of the Electrocardiogram

The electrocardiogram is a recording of the net result of the heart′s electrical activity, and there are stages in the genesis of the electrocardiogram, which are:

1. Origin of the electrical stimulus and the heart′s primary pacemaker (the sinoatrial node).
2. Spread of the electrical stimulus through the cardiac conducting system.
3. Depolarization of the atrial cells, then the ventricular cells.
4. Return of polarization (repolarization) of the previous cells in a certain sequence.

To understand the electrocardiogram, we will first discuss it on a single cardiac muscle fiber and then generalize.

The resting membrane potential is about 90 millivolts, and the reason for this is the distribution of ions on the two sides of the membrane: sodium ions are outside the cell at a concentration thirty times higher than their concentration inside the cell, while potassium ions are inside the cell at a concentration thirty times higher than their concentration outside the cell.

Since the membrane has selective permeability to potassium ions, potassium ions leak across this membrane and line up on the outer surface of the membrane, and as they exit, large negatively charged organic particles cling to them, but they cannot cross the membrane, so they line up on its inner surface. Thus the charge of the inner surface

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of the membrane is negative and the charge of the outer surface is positive, but these charges are uniform throughout the membrane, so if we placed two electrodes at two different points of the membrane, the potential difference between them would be zero because of the equality and uniformity of the charges on the outer surface of the device. The electrocardiogram consists of two parts: the part that results from the loss of polarization (depolarization) and the part that results from the return of polarization (repolarization).

[Figure: cardiac action potential. Labels: Na+ channels close; Slow Ca2+ channels open; The plateau; Slow Ca2+ channels close; Repolarization; Rapid depolarization; K+ channels close; Voltage-gated ion channels open; Refractory period; Absolute; Relative; Influx of Na+; mV (−90 to +30); Time (ms) 0–300]

The previous figure shows a single cardiac muscle fiber during four stages of depolarization and repolarization: the normal negative potential inside the fiber disappears and the membrane potential reverses, becoming slightly positive on the inside and negative on the outside.

In the figure shown, in **stage A** of depolarization, positive charges appear on the inside and negative charges on the outside, as they move from left to right; depolarization of the first half of the fiber takes place while the remaining half is still polarized, and consequently the left electrical electrode picks up a negative field while the right electrode records a positive field, and this leads to the measurement being recorded as positive.

To the right of the muscle fiber appears the potential recorded between the two electrodes as picked up by the high-speed recording device. We note that the recording has risen to its highest positive value, and that was when depolarization reached the middle of the distance along the length of the cardiac muscle fiber.

In **stage B**, depolarization extended over the whole muscle fiber and the recording on the right returned to the baseline zero line, because the two electrodes are now in areas of equal negativity, and the resulting wave is called the depolarization wave because it resulted from the spread of depolarization along the length of the muscle fiber.

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In **stage C**, the middle of the repolarization of the muscle fiber appears, with positivity returning to the outside of the fiber. At this point the left electrode is in the positive field and the right electrode is in the negative field, and as a result the recording becomes, as it appears on the right side, negative.

In **stage D**, repolarization of the muscle fiber is completed fully, with both electrodes in positive areas, and consequently no potential difference between them is recorded. Thus the recording that appeared on the right side returns to the zero level. This negative wave is called the repolarization wave because it resulted from the spread of repolarization along the length of the muscle fiber.

1-4-1 Characteristics of the Normal Electrocardiogram

The electrocardiogram consists of the P wave, the QRS complex, and the T wave, and the QRS complex often consists of separate waves, the Q, R, and S waves, but this is not always the case. The P wave results from the electrical potentials generated by depolarization of the atria before the start of atrial contraction, and the QRS complex results from the potentials generated by depolarization of the ventricles before their contraction, that is, during the propagation of the depolarization wave through the ventricles; accordingly we call the P wave and the components of the QRS complex depolarization waves. The T wave results from the potentials generated by repolarization of the ventricles, and this process normally occurs in the ventricular muscle after depolarization by a period ranging between 0.25–0.35 seconds, and the T wave is known as the repolarization wave. Before muscle contraction occurs, depolarization must spread in order to stimulate the start of the chemical processes of contraction, as the P wave occurs at the beginning of atrial contraction and the QRS complex waves occur at the beginning of ventricular contraction. The ventricles remain contracted until a few thousandths of a second after repolarization takes place, that is, after the end of the T wave. The atria repolarize about 0.15–0.20 seconds after the P wave, and this happens exactly at the moment when recording of the QRS complex begins on the electrocardiogram, and therefore the atrial repolarization wave, known as the atrial T wave, disappears because of the QRS complex, which is much larger than it, and for this reason we rarely find the atrial T wave on the electrocardiogram. The atrial repolarization wave is the T wave in the normal electrocardiogram, and usually the ventricular muscle begins to regain its polarization in some fibers about 0.20 seconds after the beginning of the depolarization wave (the QRS complex) and this is after 0.35 seconds in many other fibers. Thus the repolarization process extends over a long period reaching about 0.15

seconds, and for this reason the T wave in the normal electrocardiogram is often prolonged and extended, but the voltage of the T wave is clearly lower than the voltage of the QRS complex, and this is partly due to its extended length.

[Figure: ECG waveform. Labels: QRS Complex; R; P; Q; S; T; PR Segment; ST Segment; PR Interval; QT Interval]

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1-4-2 Normal Cardiac Rhythm

Sinus rhythm is the only normal steady rhythm. The R-R interval decreases in young people during inspiration, and this is called sinus arrhythmia. When sinus arrhythmia is pronounced it can mimic atrial arrhythmia; however, every P-QRS-T complex is normal in sinus arrhythmia and the interval between them is the only thing that changes. Sinus arrhythmia becomes less pronounced with increasing age and disappears in some cases with diabetic autonomic neuropathy, and the reason in both cases is weakness of vagus nerve function.

1-4-3 The P Wave

The P wave is normally upright (positive) in all leads except VR in sinus rhythm. When the QRS complex is predominantly downward in VL, the P wave can also be inverted. In patients with dextrocardia (right-sided heart), as in Figure (8-4), the P wave is inverted in lead I, although this pattern is often seen when the limb leads have been connected wrongly, and dextrocardia can be distinguished if the QRS complex appears predominantly downward in leads V5 and V6.

1-4-4 The PR Interval

In sinus rhythm the PR interval is constant and the normal range is (120–200 milliseconds) (3–5 small squares of the ECG paper). A PR interval of less than 120 milliseconds indicates early excitation, and a PR interval longer than 220 milliseconds is caused by first-degree block; this abnormality can be seen in healthy people.

1-4-5 The QRS Complex

The duration of the QRS complex must be less than 120 milliseconds. If it is longer than this, then either the ventricles have been depolarized from a ventricular focus and not from a supraventricular focus, or there is a conduction abnormality within the ventricles. An RSR pattern, similar to right bundle branch block (RBBB) but with a narrow QRS complex, is sometimes called partial right bundle branch block and is a normal variant.

1-4-6 The Q Wave

The normal depolarization of the septum between the ventricles from left to right causes a small septal Q wave in any of leads II, VL, or V5–V6. Septal Q waves are usually less than 3 mm deep and less than 1 mm wide. The septal Q wave appears.

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A small Q wave is also common in lead III in healthy people, and then it is always narrow but can be deeper than 3 mm.

1-5 Cardiac Arrhythmias

Cardiac arrhythmia is a common cause of illness and affects people with normal hearts as well as people who have structural or electrical abnormalities. Some arrhythmias have few hemodynamic consequences, while others may cause cardiac arrest. It is therefore important to recognize the clinical significance of cardiac rhythm disturbance. In normal sinus rhythm, the electrical impulse that causes the heart to beat arises in the sinus node and is conducted to the ventricles by the specialized cardiac conduction system. Arrhythmia is any deviation from this normal system of impulse formation and conduction. In other words, an arrhythmia occurs when the electrical impulse arises outside the sinus node or is conducted abnormally through the electrical conduction system [7]. Some of its types:

1. Atrial fibrillation (AF) and flutter
2. Supraventricular tachycardia (SVT)
3. Ventricular tachycardia (VT)
4. Ventricular fibrillation (VF)

1-5-1 Atrial Fibrillation

It is a disease of the left atrium (atrial enlargement) and is the most common arrhythmia. It occurs because of ectopic beats arising from the pulmonary veins where they enter the left atrium, with rapid and chaotic generation of beats, resulting in multiple electrical

wave pulses sweeping across the atria. They collide and extinguish and create electrical chaos in the atria.

Problems associated with AF include a significantly increased risk of stroke, heart failure caused by an uncontrolled heart rate, and unpleasant symptoms including palpitations, dizziness, and shortness of breath. On the ECG, AF has a number of key features (Figure 16-4) that make it easy to recognize:

1. P waves are replaced by fibrillation waves. These distort the baseline of the ECG in a fine, irregular pattern.

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2. The rhythm is completely irregular; in other words, there is no regular interval between QRS complexes.
3. The QRS complex is narrow, unless conduction in the ventricles is abnormal.

1-5-2 Atrial Flutter

Atrial flutter is closely related to ventricular flutter, although the mechanism is completely different from it, and the same basic conditions are associated with flutter (such as high blood pressure and coronary heart disease). The beats in atrial flutter are more regular than in atrial fibrillation.

The arrhythmia usually begins with an ectopic beat conducted abnormally through the atria. The impulse begins to circulate continuously around the ring of the tricuspid valve, advancing counterclockwise up the septum between the atria and then down the lateral wall of the right atrium. There is often a regular ratio of conducted beats.

1-5-3 Ventricular Fibrillation

There is no organized rhythm at all. The electrical activity in the ventricles is chaotic and rapid. In very rare cases, short episodes of spontaneous termination of VF occur, but in most cases the rhythm continues and causes complete loss of cardiac output and cardiac arrest. VF is often caused by severe lack of perfusion due to coronary heart disease. However, it can also occur in other forms of structural heart disease. VF can be recognized on the ECG by its chaotic appearance, as in Figure (21-4). No QRS complexes can be recognized.

1-5-4 Premature Ventricular Contractions (PVC):

Premature ventricular contractions (PVCs) are extra heartbeats that begin in one of the two lower pumping chambers of your heart (the ventricles). These extra beats disrupt the regular heart rhythm, sometimes causing a fluttering or skipped beat in your chest.

Premature ventricular contractions are common — they occur in many people.

They are also called:

- Premature ventricular complexes
- Premature ventricular beats
- Ventricular extrasystoles

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If a person has premature ventricular contractions from time to time but is in good health, there is probably no cause for concern and no need for treatment. If they have frequent premature ventricular contractions or underlying heart disease, they may need treatment.

Symptoms:

Premature ventricular contractions often cause few or no symptoms. But you may feel a strange sensation in your chest.

Causes

The causes are not always clear. Certain triggers, heart disease, or changes in the body can make the cells of the ventricles electrically unstable. Heart disease or scarring may also cause the path of the electrical impulses to change.

Premature ventricular contractions can be associated with:

- Some medications, including decongestants and antihistamines.
- Alcohol or illicit drugs.
- Increased adrenaline levels in the body, which may be caused by caffeine, tobacco, exercise, or anxiety.
- Injury to the heart muscle from coronary artery disease, congenital heart disease, high blood pressure, or heart failure.

Risk factors:

The following can increase the risk of premature ventricular contractions:

- Caffeine, tobacco, alcohol, and illicit drugs.
- High blood pressure (hypertension).
- Anxiety.

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- Heart diseases, including congenital heart disease, coronary artery disease, heart attacks, heart failure, and weak heart muscle (cardiomyopathy).

[Figure 4-1: The problem in electrical conduction in the case of PVC. Labels: Sinus node impulse; Impulse passes through AV node; Ventricular impulses; Normal heartbeat]

Complications:

The presence of frequent premature ventricular contractions, or certain patterns of them, may lead to an increased risk of developing heart rhythm problems (arrhythmias) or weakened heart muscle (cardiomyopathy).

Rarely, when accompanied by heart disease, frequent premature contractions can lead to a chaotic and dangerous heart rhythm and possibly sudden cardiac death.

How are PVCs diagnosed?

Premature ventricular contractions can be difficult to diagnose because they occur at unexpected intervals. In most cases, it is difficult for the doctor to detect a premature ventricular contraction during a routine procedure unless the patient has one during the examination or has other signs of structural heart problems. In individuals who have no known heart disease, PVCs are often detected incidentally during a routine electrocardiogram (ECG). In patients who suffer from known heart disease, PVCs may be detected during other diagnostic tests for that condition.

A patient who suffers from symptoms of premature ventricular contraction and who consults a cardiologist will undergo an electrocardiogram, which is the standard test for diagnosing premature ventricular contraction. It is relatively inexpensive, non-surgical, and accurate

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and precise. A Holter monitor can be used, which is a portable ECG device that provides 24 or 48 hours of continuous recording, for patients who suffer from a premature ventricular contraction at least once a day. The information obtained from the Holter monitor is very useful in determining the frequency of premature ventricular contractions and linking symptoms such as fainting or dizziness with irregular heartbeats.

The American College of Cardiology recommends an exercise test — measuring heart rate while walking on a treadmill — for some patients who suffer from premature ventricular contractions who were diagnosed by electrocardiogram.

Major heart centers such as Cleveland Clinic also have other high-level specialized tests available for patients who suffer from premature ventricular contractions and who cannot undergo exercise tests or who are at high risk of developing more complex heart problems. These tests may include cardiac magnetic resonance imaging, cardiac computed tomography, computed tomography of myocardial perfusion, coronary angiography, or electrophysiology testing.

[Figure 4-2: Comparison between the normal signal and the PVC signal. Panels: Normal Beat; PVC Beat; axes: Amplitude (mv), Time (s)]

1-5-5 Left Bundle Branch Block (LBBB):

Left bundle branch block (LBBB): is a condition in which there is a delay or blockage along the pathway through which electrical impulses travel to make your heart beat. Sometimes it is difficult for your heart to pump blood efficiently through your body.

The delay or blockage can occur in the pathway that sends electrical impulses either to the left or right side of the lower chambers (ventricles) of your heart.

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[Figure: Heart block. Labels: Sinus node; Atrioventricular (AV) node; Right atrium; Left ventricle; Right ventricle; Heart block. © Mayo Foundation for Medical Education and Research. All rights reserved.]

Symptoms:

In most people, bundle block does not cause symptoms. Some people with this condition do not know that they have a bundle branch block.

Rarely, signs and symptoms include fainting or feeling as if you are going to faint.

Causes

Usually, electrical impulses inside the heart muscle cause the heart to beat (contract). These impulses travel along a pathway, including two branches called the right and left bundles. If one or both of these branches are damaged — because of a heart attack, for example — the electrical impulses can be blocked, and your heart beats abnormally.

The cause of bundle branch block can differ depending on whether the left or the right bundle branch is affected. Sometimes, there is no known cause.

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Accordingly, the stretching of heart tissue can lead to problems in conduction between the heart muscles. The main cause of LBBB is dilated cardiomyopathy. In particular, enlargement of the left ventricle leads to stretching and separation of Purkinje fibers.

The causes can include:

- Heart attack (myocardial infarction).
- High blood pressure (hypertension).
- Infection of the heart muscle with bacteria or a virus (myocarditis).
- Enlargement, hardening, or weakening of the heart muscle (cardiomyopathy).

And (myocardial infarction) is among the most common causes.

Risk factors:

Risk factors for bundle block include the following:

- Advancing age. Bundle block is more common in older people than in young people.
- Underlying health problems. High blood pressure or heart disease increases your risk of bundle block.

Complications:

If both the right and left bundles are blocked, the main complication is a complete blockage of the electrical signals from the upper chambers to the lower chambers of the heart. The lack of signals can slow the heart rate, leading to fainting, irregular heartbeats, and other serious complications.

Because bundle block affects the electrical activity of your heart, it may sometimes complicate the accurate diagnosis of other heart diseases, especially heart attacks, and lead to delay in the proper management of those problems.

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The First Study:

Zubair, M.; Kim, J.; Yoon, C. An Automated ECG beat classification system using convolutional neural networks. (2019).

An automatic classification system based on deep learning (DL) technology was proposed for monitoring and detecting heart diseases (CD). The proposed DL structure is divided into unsupervised Deep Auto-Encoders (DAEs) to learn the features, and Deep Neural Networks (DNNs) as a classifier. This is done using samples of ECG signals from the MIT-BIH database, where 7 classes of diseases are classified, the most important being premature atrial contraction, left bundle branch inflammation, normal heartbeat, and tachycardia. Processing is done by dividing the signal into more than one separate sample, and then removing the noise from each of them through a number of filters. 39,000 samples were used, divided among the cases mentioned previously, in a certain proportion.

The results found that the proposed method achieved promising performance: an accuracy of 99.73%, and a sensitivity of 91.20% and 91.80% (F1-Score).

The Second Study:

Ullah, A.; Anwar, M.S.; Bilal.M.; Majid, R.M.; Classification of Arrhythmia by Using Deep Learning with 2-D ECG Spectral Image Representation. (2020).

Careful study of electrocardiogram signals is extremely important for the accurate diagnosis of acute and chronic heart diseases. In this study, we propose a two-dimensional convolutional neural network (2D-CNN) model to classify electrocardiogram signals into eight categories; that is, normal beat, premature ventricular contraction beats, right bundle branch block beats, left bundle branch block beats, premature atrial contraction beats, ventricular flutter wave beats. The one-dimensional ECG time-series signals are converted into two-dimensional spectral images through a short-time Fourier transform. The 2D CNN model, consisting of four convolutional layers and four pooling

layers, was designed to extract robust features from the ECG spectrograms. Cardiac signals from the BIH-MIT database are used, but the study did not mention the amount of data used. A state-of-the-art average classification accuracy of 99.11% was achieved.

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The Third Study

Deb, Pratik et al. "Detection of Abnormal Electrocardiogram (ECG) Using Wavelet Decomposition and Support Vector Machine (SVM)." 2019 1st international Conference on Advances in science, Engineering and Robotics Technology (ICASERT) (2019): 1-8.

An automatic diagnostic system based on an SVM classifier was proposed with the aim of classifying the ECG signal as a normal or abnormal signal. The classifier classifies signals that are considered indicative of: angina, heart block, cardiomyopathy, heart failure, arrhythmia, hypertrophy of the heart muscle, myocardial infarction, myocarditis, and valvular heart diseases. All of them were considered abnormal signals, and anything else would be classified as normal. By extracting the features of skewness, kurtosis, standard deviation of the details, and approximation coefficients of wavelets of order 5, the SVM classifier was trained and tested on 36 samples from the MIT-BIH Normal Sinus Rhythm Database and 36 from the PTB database, and thus classification accuracies of 98.61%, 97.37%, and 97.22% respectively were reached. The size of the data adopted by the study is relatively small.

The Fourth Study

Dohare, Ashok Kumar. "Classification of Normal and Abnormal ECG Using Composite Lead Features with SVM Classifier." (2018).

Normal and abnormal electrocardiogram signals were classified using composite lead parameters. The composite lead consists of 12 leads of the ECG signal. The following features were extracted: clinical wave intervals, amplitudes, and statistical parameters such as area, standard deviation, and mean, of the composite lead, to classify the normal and abnormal electrocardiogram signal using a support vector machine (SVM) classifier. In this study, the (PTB) database was taken, and 60 normal cases (N) from 52 persons and 60 abnormal cases (A) from 148 persons were considered. The performance accuracy was 83.33% (60/58).

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Chapter Two

(Software and Techniques Used)

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2-1 Artificial Intelligence

Artificial intelligence is a branch of computer science concerned with developing systems and programs capable of performing tasks that are usually considered the preserve of human intelligence. Artificial intelligence aims to create and develop technological systems that learn from data and interact with the environment and adapt to achieve specific goals in an intelligent way.

Artificial intelligence has developed over time and witnessed tremendous progress. In the past, the focus of artificial intelligence was on creating strictly programmed models and rules to simulate human behavior. But with the advancement of technology and the increase in processing and data storage capacity, intelligent systems have become able to analyze huge amounts of data and extract patterns and rules of behavior from them.

[Figure: nested circles — Artificial Intelligence (outer), Machine Learning (middle), Deep Learning (inner)]

Machine learning is a branch of artificial intelligence that focuses on developing models and systems able to learn and improve their performance automatically from data. Machine learning models are trained on a set of data and extract the statistical models and rules necessary to make decisions or perform tasks better. Deep learning emerges from a special type of machine learning, and it is considered a sub-field of machine learning. Deep learning depends on deep artificial neural networks, which

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usually have a multi-layered structure consisting of successive layers of computational units connected to one another. Deep learning is used in image recognition, machine translation, text generation, and other applications that require deep understanding of complex data.

2-2 The Importance of Artificial Intelligence

Artificial intelligence is characterized by exceptional importance in multiple fields, as it is considered an advanced technological sector aiming to design technical systems capable of performing tasks that require thinking and decision-making in a manner similar to humans. Artificial intelligence includes multiple branches such as machine learning, intelligent analysis, machine learning, smart robots, automated conversation, classification, and prediction. The importance of artificial intelligence is reflected in many important aspects:

- First, artificial intelligence contributes to improving productivity and increasing efficiency in various sectors. Through control of human errors and performing repetitive tasks with high accuracy and speed, smart robots can enhance production processes and achieve cost reductions.
- Second, artificial intelligence contributes to developing the health sector, as artificial intelligence systems can improve patient care and disease diagnosis. By analyzing medical data, medical history, and symptoms, correct diagnoses can be identified and appropriate treatments directed. Moreover, surgical robots can be used in performing precise and effective surgical operations.
- Third, artificial intelligence contributes to improving the user experience in various fields. Through machine learning applications, personalized recommendations and content customization can be provided according to users' interests. In addition, automated conversation systems can provide immediate and responsive support to customers' needs in the services sector.
- Fourth, artificial intelligence can predict future events and help in making decisions. Big data analysis and machine learning can reveal patterns and trends invisible to humans, and this makes companies and institutions able to predict demand for products and improve marketing and planning strategies.

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Artificial intelligence carries many opportunities and challenges in our current era. By using advanced technology and intelligent analysis, major improvements can be achieved in various fields such as productivity, health, user experience, and decision-making. However, working in this field requires good attention to security and privacy issues, in addition to directing efforts toward developing artificial intelligence technology in an ethical and sustainable manner.

2-3 Drawbacks and Limits of Artificial Intelligence

Artificial intelligence, as a modern and advanced technology, faces a number of drawbacks and limits that touch on its technical and ethical aspects. Researchers, engineers, and decision-makers must realize and address these challenges to achieve sustainable progress in the field of artificial intelligence. This analytical article begins by focusing on the lack of human understanding and interpretation that cannot be achieved by artificial intelligence. Despite the development of artificial intelligence in the fields of machine learning and data analysis, it lacks the ability to comprehensively understand context, feelings, and emotions as a human does. This deficiency can lead to inaccurate interpretations of information and inappropriate decisions.

In addition, security and privacy issues are among the most important challenges facing artificial intelligence. Training and operating deep models requires large amounts of personal data, which increases the risk of violating users' privacy. Moreover, concerns related to cyber breaches and exploitation in fraud and espionage operations occupy an important place in public debate. Ethical issues also play an important role in the context of artificial intelligence, as it may suffer from bias and unfairness. If artificial intelligence models are trained on biased or unbalanced data in terms of diversity, this may lead to decisions that favor a certain group or discriminate against another. Consequently, artificial intelligence models should be designed and trained in ways that aim to achieve fairness and balance in the results.

Dependability and errors are another challenge that should be addressed in the context of artificial intelligence. Artificial intelligence relies on data for learning and training, and if this data is distorted or incomplete, this may lead to inaccurate results and wrong decisions. In addition, artificial intelligence may face difficulty in dealing with new or unfamiliar situations that it has not been trained on beforehand.

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2-4 What Is the Concept of Machine Learning

Machine learning is a dynamic and rapidly evolving field of study in artificial intelligence. It involves a multidisciplinary approach that focuses on developing, analyzing, and implementing statistical algorithms able to learn independently using a set of data, enabling systems to distinguish patterns, predict, and perform complex tasks automatically and without the need for direct instructions.

The main goal of machine learning is to provide computer systems with the ability to learn from past experiences or observations and generalize that knowledge to deal with new data, which is called unseen data. By taking advantage of statistical techniques, machine learning algorithms extract meaningful insights and patterns from huge amounts of structured and unstructured data, enabling systems to make decisions, recognize patterns, and perform tasks with remarkable accuracy. In recent years, the emergence of artificial neural networks (ANNs) has revolutionized the field of machine learning. Artificial neural networks, inspired by the complex network of interconnected nerve cells in the human brain, have proven highly effective in solving a wide range of complex problems. Through layers of interconnected nodes, artificial neural networks can capture complex relationships and nonlinear dependencies within data, enabling them to outperform many traditional methods in terms of performance and accuracy. The success of machine learning has led to its widespread adoption in various industries and fields. It has applications in computer vision, natural language processing, speech recognition, recommendation systems, self-driving vehicles, medical diagnosis, financial forecasting, and many other fields. As machine learning techniques continue to advance and evolve, they have the potential to revolutionize many aspects of our daily lives, making machines more intelligent, adaptive, and able to help humans in increasingly sophisticated ways.

2-5 Applications of Machine Learning

Machine learning has made significant contributions across many real-world applications, leading to transformative changes in various industries.

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One of the areas in which machine learning has had a profound impact is the field of health care. By analyzing huge amounts of medical records, laboratory results, and imaging data, machine learning algorithms help doctors make accurate diagnoses and

develop effective treatment plans. For example, a pioneering study conducted showed how a machine learning model outperformed dermatologists in identifying skin cancer by analyzing images. The ability of machine learning to detect subtle patterns and anomalies in medical data enhances patient care and improves health outcomes.

In the financial sector, machine learning algorithms have become indispensable tools for fraud detection and risk assessment. By analyzing comprehensive financial data, machine learning models can identify fraudulent transactions, anomalies, and suspicious patterns that might pass unnoticed by traditional methods. The effectiveness of machine learning in detecting fraud on credit cards has been demonstrated, leading to reduced financial losses for consumers and institutions. The ability of machine learning to process huge amounts of data and make quick decisions enhances the stability and security of financial systems.

Machine learning has revolutionized marketing and predicting customer behavior. Machine learning algorithms analyze huge amounts of customer data to gain insight into preferences, behavior patterns, and purchasing habits. This allows companies to personalize marketing campaigns, develop recommendation systems, identify potential customers, and improve customer segmentation. Machine learning enables companies to make data-driven decisions, improve marketing strategies, and enhance customer experiences in a competitive market.

In the field of transportation, machine learning is revolutionizing the industry, especially with the development of self-driving vehicles. Machine learning algorithms process data from various sensors, such as cameras and laser sensing technology, to make real-time decisions, enabling self-driving cars to navigate roads safely and efficiently. Some companies have made remarkable progress in this field, as their self-driving vehicles have covered millions of miles on public roads. The ability of machine learning to analyze complex environmental data and learn from real-world scenarios is reshaping the future of transportation.

Machine learning algorithms have also played a pivotal role in developing natural language processing (NLP) capabilities. Through advanced machine learning models, machines can understand and generate human language, enabling applications such as language translation,

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sentiment analysis, and voice assistants. The development of advanced architectures such as Google's transformer has pushed neural language programming to new horizons, allowing more precise and accurate interactions between humans and machines.

These examples illustrate how machine learning is transforming various industries, improving operations, improving the decision-making process, and enhancing user experiences. As machine learning continues to evolve and unleash new possibilities, its impact on the real world is poised to expand even further, leading to stimulating innovation and shaping the way we interact with technology in our daily lives.

2-6 Machine Learning

As we mentioned, machine learning aims to develop models and systems able to learn and adapt through analyzing data and acquiring knowledge automatically. Supervised learning and unsupervised learning are considered two main modes in the field of machine learning, and they differ in the way in which models are trained and knowledge is acquired.

2-6-1 Supervised Learning

In supervised learning, models are trained using a set of data containing input and output pairs that are fully known in advance. The training data is provided, where the correct answer is provided for each input. The model analyzes the data and extracts the relationships and patterns that exist between inputs and outputs. After good training, the model becomes able to predict the required outputs for new inputs that are not known in advance.

Linear Regression Algorithm

Linear regression is a classic algorithm used for regression tasks. It predicts a continuous target variable based on input features. It fits the line that best represents the relationship between the independent variables and the dependent variable. The model estimates the line's coefficients through a process called ordinary least squares. Linear regression finds applications in various fields, such as economics and finance.

This algorithm aims to determine the linear relationship between input and output. The goal is to find an equivalence line that comes as close as possible to the available data. This is achieved by a simple linear fit that represents the relationship between the independent variables and the target variable. This line expresses the relationship with the linear regression equation:

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y = mx + b

where:

y is the target variable.

x is the independent variable.

m is the slope coefficient, which represents the amount of change in the target value relative to the independent variable.

b is the intercept, and it expresses the expected value of the target variable when the independent variable equals zero.

[Figure: scatter of points with a fitted regression line. Axes: X1, X2]

The algorithm works by minimizing the differences (errors) between the actual target values and the expected values estimated by the linear model. The least squares principle is used to estimate the best value for the slope coefficient and the intercept.

Advantages of the Linear Regression Algorithm

Simplicity of implementation: Linear regression is considered one of the simplest statistical and analytical methods, and it can be easily understood and implemented. Beginners in the field of statistical analysis and machine learning can benefit from it and apply it easily.

Finding the linear relationship: Linear regression helps in determining the linear relationship between the independent variables and the dependent variable. It can be used to analyze the effect of the independent variables on the dependent variable and to estimate the strength of the relationship between them.

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Prediction and forecasting: Linear regression can be used to predict values of the dependent variable based on the values of the independent variables. This ability can be used in sales forecasts, economic growth forecasts, profit estimates, and many other fields.

Impact evaluation and experiments: Linear regression can be used to estimate the effect of the independent variables on the dependent variable. It can be used in scientific experiments and research to analyze the effect and evaluate the changes that occur as a result of intervention in the independent variables.

Disadvantages of the Linear Regression Algorithm

Linearity assumptions: Linear regression assumes the existence of a linear relationship between the variables. This assumption may not be accurate in many real cases where the relationships are nonlinear. Ignoring nonlinear relationships may lead to incorrect predictions.

Dealing with outliers: Outliers can have a large effect on linear regression estimates and can lead to distortion of the results. Dealing with outliers may require additional procedures such as transforming variables or excluding unusual values.

Dependence on training data: The performance of linear regression depends on the quality and representativeness of the training set used. If the data is not representative of the real situation or contains distortions or missing values, this may have a negative effect on the results.

Sensitivity to multicollinearity: In the case of multicollinearity in the data, linear regression may have difficulty dealing with it correctly. Multicollinearity may lead to distortion of the estimates and a decrease in model accuracy.

Uses of the Linear Regression Algorithm

First, linear regression is used to analyze the relationship between variables. It can be used to understand the correlation between two or more variables in a given context. For example, we can use it to understand the relationship between students' grades and the number of hours of

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study, or between the level of advertising spending and the level of sales. By means of linear regression, we become able to clarify the correlation between these variables and predict future values.

The second use comes in the field of prediction and forecasting, where linear regression can be used to forecast future values. We can use it in economic growth forecasts or stock market forecasts, where the linear model calculates the relationship between the

independent variables and the dependent variable to achieve accurate forecasts of future events. In addition, linear regression can be used for impact analysis and experiments. In this case, intervention is applied to the independent variables to measure their effect on the dependent variable. We can use linear regression to evaluate the effect of a training program on employee performance or the effect of a medical treatment on patients' condition.

The fourth case comes in the field of description and exploratory analysis, where linear regression can be used to analyze the factors affecting the dependent variable and describe the relationship between them. We can use it to understand the factors affecting customer satisfaction or to estimate the strength of the relationship between multiple factors in exploratory studies.

Finally, linear regression can be used to tune models and improve their performance. We can use it to tune coefficients and select influential variables to achieve better performance of statistical models and achieve more accurate predictions.

Decision Tree Algorithm

Decision trees are tree-based models that make decisions based on feature values to predict the target variable. They split the data repeatedly into subsets, taking different features into account at each node, until a stopping criterion is met. Decision trees provide interpretable models and are widely used in classification and regression tasks. They find applications in fields such as customer segmentation and fraud detection. The decision tree algorithm depends on dividing the data into subsets based on the different attributes of the available data. The process consists of choosing the best attribute for splitting and building a tree consisting of nodes and branches. This process is repeated until a certain goal is achieved, such as accurate classification of the data or prediction of target values.

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[Figure: example decision tree with Arabic labels — Forecasts; Sunny; Windy [as OCR'd: "عالم"]; Humidity; Temperature; Wind; High; Moderate; Yes; No]

Advantages of the Decision Tree Algorithm

Ease of interpretation: The decision tree is considered one of the tools that are easy to interpret and visualize. Users can easily understand the decisions and conditions used in the tree, as it can be represented in a simple and logical form.

Ability to deal with mixed data: The decision tree allows dealing with mixed data, and it can deal with a diverse set of variables such as numbers, texts, and categories.

Effectiveness in prediction: The decision tree is distinguished by its ability to predict future outcomes effectively. It can be used to predict probabilities and make informed decisions.

Ability to deal with big data: The decision tree can deal with big and complex data. The performance of the tree can be improved through techniques such as partitioning and clustering.

Disadvantages of the Decision Tree Algorithm

Exposure to the problem of increased depth: The decision tree may be exposed to the problem of increased depth as a result of the increase in the number of decisions and conditions. This may lead to the complexity of the tree and the difficulty of understanding and interpreting it.

May be prone to bias: The decision tree may be prone to the problem of overfitting, which is a case that occurs when the tree is very adapted to the training data used and does not work well on new data.

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Dealing with continuous variables: The decision tree is ineffective in dealing with overlapping continuous variables. It is necessary to use special techniques such as clustering and partitioning to deal with this type of variable.

Uses of the Decision Tree Algorithm

The decision tree is used in many different cases:

Feature analysis and variable selection: The decision tree can be used to analyze features and determine the variables with the greatest effect on the desired outcomes. This allows a comprehensive understanding of the relationships between variables and ranking them by importance.

Data classification: The decision tree can be used to classify data into different categories. It can be used in binary classifications (such as "yes" or "no") and multi-class classifications (such as "low," "medium," or "high").

Product recommendations and marketing: The decision tree can be used to provide informed product recommendations and marketing. It can help determine the best marketing strategy for customers based on their attributes and needs.

Disease diagnosis and medical decision-making: The decision tree can be used in the medical field to diagnose diseases and make medical decisions. It can analyze the patient's symptoms and medical history and provide recommendations for appropriate diagnosis and treatment.

Risk analysis and financial decision-making: The decision tree can be used to analyze risks and make financial decisions. It can estimate potential risks, analyze influencing factors, and provide recommendations for risk management and wise financial decisions.

Customer analysis and marketing decision-making: The decision tree can be used to analyze customer-related data and make effective marketing decisions. It can determine the factors influencing purchase decisions and provide recommendations for targeted marketing strategies.

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Random Forest Algorithm

Random forest works by creating a set of decision trees, where each tree is trained on a random subset of the data and a subset of the features. During training, each tree makes predictions independently, and the final prediction is determined by aggregating the results from all the trees. This aggregation can be done through majority voting for classification tasks or by calculating the average for regression tasks. The random forest algorithm depends on forming a set of simple decision trees. Each sub-tree is trained on a random sample of the data, and the final decision is made through the majority vote of the sub-trees. The sub-attributes used in each sub-tree are selected randomly and independently.

Advantages of the Random Forest Algorithm

Predictive power: Random forest is considered strong in prediction and classification, as the predictions of several independent decision trees are collected to obtain a final classification that is more powerful and accurate.

Dealing with big data: Random forest works well with large datasets, as the task can be divided into several independent decision trees and trained in parallel, which speeds up the classification process.

Reducing bias: Since random forest depends on a set of independent decision trees, it reduces the bias related to any individual decision tree and promotes fairness and objectivity in classification.

Dealing with variables: Random forest allows dealing with a wide range of variables, including numerical and categorical variables, without the need to transform or prepare them in advance.

Disadvantages of the Random Forest Algorithm

Computational complexity: It requires building and training many decision trees, which increases computational complexity and the consumption of time and computing resources.

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Stagnation: Random forest may suffer from stagnation in classification when it faces data with little diversity or in the case of variables with a strong influence on the classification that overpower the others.

Difficulty of interpretation: It may be difficult to interpret the reason for a certain classification or understand the relationship between variables in random forest because of the multiplicity and complexity of the trees.

Need to tune parameters: Using random forest requires tuning some parameters such as the number of trees and the depth of each decision tree, and this needs experimentation and optimization to reach optimal performance.

Uses of the Random Forest Algorithm

This algorithm helps in many fields, the most important of which are:

The random forest algorithm is used in many suitable use cases. One of these cases is data classification, where random forest can be used to classify images and texts and determine patterns in numerical data. The algorithm creates a set of random trees that generate multiple sub-classifications, and then aggregates them to reach an accurate final classification. In addition, random forest can be used in providing personalized

recommendations. It can analyze the user's records and preferences and provide customized and appropriate recommendations for each individual. This can be applied in the field of online shopping, where it can deliver suitable products to each user based on their preferences and browsing history.

The ability of random forest to detect fraud is also one of the suitable cases for use. It can analyze unusual patterns and behaviors of users and identify potential fraudulent activities. This feature can be used in financial services and insurance to protect institutions and customers from fraud. In addition, random forest is used in analyzing medical data and diagnosing diseases. It can analyze patient records and clinical information to provide an accurate diagnosis and treatment recommendations for doctors, which helps improve the quality of health care.

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Finally, random forest is used in analyzing engagement on social media. It can classify content, analyze patterns, and extract opinion and sentiment from users, which helps marketers and companies understand audience interaction and improve their marketing strategies.

Support Vector Machine (SVM) Algorithm

A supervised machine learning algorithm capable of performing classification and regression and even detecting anomalies. The Support Vector Machine (SVM) works by drawing a straight line between two classes called the margin; the data points that fall to the right or left of the margin are classified within one class, then the algorithm determines the line that separates the two classes and stays as far as possible from the nearest samples. For example: if we want to classify data points into two different classes (circles and triangles), we start by drawing a random plane, then we check the distance between the plane and the nearest data points from each class; the data points closest to the plane are known as support vectors, and the plane is drawn based on them, and the optimal plane represents the maximum distance from all support vectors.

[Figure: SVM with two classes, the separating line and margins. Labels: Support vector (twice); Axes: X1, X2]

Advantages of the Support Vector Machine Algorithm:

Effective with high-dimensional data: This algorithm is particularly effective when dealing with datasets that contain a large number of features. It can deal with complex relationships and nonlinear decision boundaries by mapping the data to higher-dimensional spaces where separation is easier.

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Robust against overfitting: SVM is less prone to overfitting compared to other algorithms such as decision trees. It finds the balance between maximizing the margin and minimizing classification error, which helps in generalizing to unseen data well.

Effective on small and medium-sized datasets: The algorithm works well when the size of the dataset is medium. It can deal with datasets that contain a few thousand samples efficiently and provide accurate classification results.

Disadvantages of the Support Vector Machine Algorithm:

Sensitivity to parameter tuning: The algorithm contains many parameters that need to be tuned correctly to achieve optimal performance. Determining the appropriate values for these parameters may be a difficult matter and may require cross-validation or grid search techniques.

Difficulty in dealing with datasets that contain noise: If the dataset contains a large amount of noise or overlapping classes, the algorithm may face difficulty in finding accurate decision boundaries. Preprocessing techniques such as feature selection or noise reduction methods may be necessary to improve performance.

Uses of the Support Vector Machine Algorithm:

Text and sentiment analysis: It is widely used for text classification tasks, such as sentiment analysis, spam detection, and document classification. It can effectively deal with high-dimensional text data and capture complex relationships between words.

Image recognition: It has been successfully applied to image recognition tasks, including object detection, face recognition, and handwritten digit recognition. By converting images into feature vectors, it can classify and identify objects within images.

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Financial analysis: It is used in financial markets for tasks such as stock price prediction, credit scoring, and fraud detection. It can analyze complex financial data and identify patterns and anomalies that can help in decision-making.

2-6-2 Unsupervised Learning

In unsupervised learning, no prior guiding information is provided to the model in the training process. Instead, the model is exposed to a set of unlabeled data and it is arranged and clustered based on the patterns and similarities found in the data. The goal is to discover a hidden structure or natural clusters in the data without external help.

K-Means Algorithm

It uses a complete set of data points, divided into several classes to predict the classification of a new sample point. One of the drawbacks of the KNN algorithm is that it works by storing the whole dataset and comparing the new points with the existing points, which leads to an increase in the training set, and thus an increase in the estimation time in proportion to the number of training points. For example: if we have a number of different points divided into two classes, the first class representing outstanding students and the second class representing non-outstanding students, and we have a new point for which we want to know which class it belongs to by examining its nearest 5 points, for example, where K=5 (and it is preferable for it to be an odd number), to determine which class it belongs to, the first being outstanding students and the second non-outstanding students, and based on that the decision is made.

Advantages of the K-Means Algorithm

Scalability: It is a fast and relatively scalable algorithm, which makes it suitable for large datasets containing a large number of samples. It can handle millions of data points efficiently.

Simplicity and ease of implementation: K-means is a straightforward algorithm with a simple concept. It is easy to understand and implement, even for individuals with limited knowledge of machine learning.

Interpretability: The algorithm's results are intuitive and interpretable. Each data point is assigned to one of the K clusters, which makes it easy to understand the clustering and analyze the characteristics of each cluster.

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[Figure: three clusters of points (blue, red, green)]

Disadvantages of the K-Means Algorithm

Determining the optimal number of clusters: Determining the appropriate number of clusters (K) is a difficult task. Determining an incorrect value of K can lead to clustering results that are below the expected level or misleading.

Dealing with nonlinear or irregularly shaped clusters: K-means assumes that the clusters are spherical and have equal variance. It may find difficulty in identifying nonlinear or irregularly shaped clusters.

Sensitivity to outliers: This algorithm is considered sensitive to outliers because they can significantly affect the position of centroids. Outliers can distort clustering results and lead to inaccurate clusters. Preprocessing techniques such as detecting and removing outliers may be necessary to mitigate this problem.

Uses of the K-Means Algorithm

Customer segmentation: It can be used to group customers based on demographic composition, purchasing behavior, or their particular preferences. This information can be valuable for targeted marketing, personalized recommendations, and customer relationship management.

Anomaly detection: It can be used to detect anomalies by identifying data points that deviate significantly from the normal clusters. Unusual patterns or outliers can be discovered by considering them anomalies.

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Document clustering: It can cluster documents based on their content or similarity. This can be useful for organizing large collections of documents, topic modeling, information retrieval, and text extraction.

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Chapter Three

(Working Mechanism)

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3-1 Working Algorithm

[Figure: flowchart — Obtaining the data and reviewing it → Initial processing of the data → Feature extraction → Processing the extracted features → Training and testing different ML algorithms → Comparing the results and choosing the best algorithm → Displaying and linking the work with an interactive interface]

3-1-1 Obtaining and Reviewing the Data:

Artificial intelligence and machine learning algorithms depend fundamentally and heavily on data, which is used to train them on a particular issue, which is classification in our case here, where they are trained on a certain amount of data and tested on another amount, and thus acquire knowledge and the ability to make a certain decision.

The MIT-BHA database was relied upon, which contains samples whose number reaches 482 and includes the following cases:

The normal case (Normal)

1- Premature Ventricular Contraction (PVC)

2- Left Bundle Branch Block (LBBB)

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Each sample of the data is a signal consisting of 13 cycles, with 3600 samples within the complete signal.

[Figure: plot of the full ECG signal; vertical axis 900–1250, horizontal axis 0–3500]

The number of samples for each case in the database, as shown in the table:

Coding	Number of samples	Case
0	246	Normal
1	103	PVC
2	133	LBBB

3-1-2 Data Processing

Given the small number of data relative to the number of samples within each signal, work was done to augment the data by dividing each signal into a number of separate signals, and the division was done as follows:

First: finding the highest peak in the signal, which is the peak of the wave (R) in one of the signal's cycles. We obtain the following result.

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[Figure: the signal with the highest peak marked (legend: max_value)]

Second: for the rest of the (R) peaks, we take the peaks that make up 90% of the (R) peak detected in the previous step, and the result is:

[Figure: the signal with all R peaks marked (legend: max_value)]

Third: at each peak detected in the previous step, we take 100 steps backward and 200 steps forward. Thus we obtain one cycle of the signal.

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[Figure: a single cycle with 100 samples before and 200 samples after the R peak]

And thus the data we have is enlarged to become 1880 samples, and its number after the increase becomes as follows:

Coding	Number of samples	Case
0	984	Normal

Coding	Number of samples	Case
1	412	PVC
2	532	LBBB

Then a median filter is passed over each new signal to remove the accompanying noise.

3-1-3 Feature Extraction

We prepared an algorithm named (find_PQRST) to extract several features that we found to be valuable in classifying the studied cases, and these features are:

1- The P wave and its location.

2- The Q wave and its location.

3- The R wave and its location.

4- The S wave and its location.

5- The T wave and its location.

6- The P-Q distance

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7- The P-R distance

8- The Q-R distance

9- The R-S distance

10- The S-T distance

Feature	Significance
P	The P wave represents atrial depolarization, which leads to atrial contraction
Q	Q waves represent the initial phase of ventricular depolarization
R	The R wave represents the early ventricular depolarization process

Feature	Significance
S	The S wave indicates the final depolarization of the ventricles
T	The T wave represents repolarization of the ventricles.
P-R	Represents electrical depolarization of the atria of the heart and thus the regularity of arrival of the electrical signal to the ventricles.
T-S	The ST segment represents the time interval between ventricular depolarization and repolarization

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3-1-4 Feature Extraction Algorithm:

- Extracting the R wave:

The R wave is the highest peak present in the signal, so the signal is read in full, and every value that is large relative to the two neighbors before it and after it (a peak) is placed in a list named _temp, and at the end the largest value is taken, which is the (R) wave, and thus we have determined the value of R and its location.

- Extracting the S wave:

The S wave is a negative wave relative to the R, P, and T waves, so a range in which it is likely to be present is determined, and it is here the range between the R wave and the end of the signal.

The specified range is read and every low value relative to the neighbors before and after it is determined, and then the lowest value among the detected values is chosen, and thus we have also obtained the value of the S wave and its location.

- Extracting the Q wave:

In the same way as extracting the S wave, but with a different range, as the expected range for the presence of the Q signal is between the start of the signal and the R wave.

- Extracting the P wave:

The P wave is the first large peak in the cardiac signal before R, and it has a positive value; it is the opposite of the Q wave and in the same direction as the R wave but of a smaller value.

The extraction process is the same as the extraction of R but with a smaller range, which is between the start of the signal up to the Q wave.

- Extracting the T wave:

The T wave is the peak that follows the S wave, so it is computed in the same way as the P wave but with the range from the S wave to the end of the signal.

- Extracting the distances between the waves:

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It is a simple process; after extracting each wave and its location, the distances between the locations between the recorded points are calculated, and thus we obtain the required distances.

3-1-5 Processing the Extracted Features

Then the extracted features were arranged within a new database, while keeping the coding of what each set of features of signals represents.

	m	R	Ri	Q	Qi	P	Pi	S	Si	T	Ti	label
0	967.215153	1106.711735	96	951.491666	83	994.603084	43	948.618924	182	975.074989	222	0
1	965.240000	1072.304844	96	940.995534	85	985.159906	43	932.262386	189	975.640723	223	0
2	970.257825	1112.633331	96	949.071140	84	994.210729	44	949.250450	190	980.526765	229	0
3	971.500000	1091.101253	96	940.468808	84	995.408438	41	952.795059	181	982.879828	232	0
4	955.270000	1080.015563	96	931.390023	82	977.388117	43	933.301969	189	963.421633	235	0

At this stage, the machine learning algorithms are prepared and several types of them are tried, and the algorithm with the best results will be adopted.

For training, the input and output of the algorithm must be determined, since supervised learning algorithms are used. The input is determined, which is the extracted features, and the output is the coding for each case.

[Figure: the same table with the feature columns marked "Input" (الدخل) and the label column marked "Output" (الخرج)]

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The data is also divided between training, testing, and trial:

Case name	Total number of samples	Training data 60%	Test data 30%	Trial 10%
Normal	984	590	296	98
PVC	412	248	123	41
LBBB	532	319	160	53

3-1-6 Training and Testing Different ML Algorithms:

Four different algorithms were trained and tested for the classification of the adopted cases, in order to adopt the algorithm with the best results in the next stage.

The First Algorithm:

(Support Vector Machine) is a linear model for classification and regression problems. It can solve linear and nonlinear problems and works well with many practical problems. The idea of SVM is simple: the algorithm creates a line or hyperplane that separates the data into classes.

SVM tries to determine the decision boundaries in a way that makes the separation between the two classes as wide as possible. It is clear that this data is not linearly separable, and in our case we cannot draw a straight line that can classify this data. However, this data can be transformed into linearly separable data with higher dimensions. Let us add another dimension and call it the z-axis. We let the coordinates on the z-axis be governed by the constraint $z = x^2 + y^2$.

The Second Algorithm:

The KNeighbors Classifier algorithm was used. It is a type of supervised ML algorithm that can be used for both classification and regression predictive problems. However, it is mainly used for classification predictive problems in industry. The following two properties define KNN well

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Lazy learning algorithm – KNN is a lazy learning algorithm because it has no specialized training phase and uses all the data for training while classifying.

Non-parametric learning algorithm – KNN is also a non-parametric learning algorithm because it assumes nothing about the underlying data.

The K-nearest neighbors (KNN) algorithm uses "feature similarity" to predict the values of new data points, which also means that the new data point will be assigned a value based on how closely it matches the points in the training set. We can understand its work with the help of the following steps:

Step 1 To implement any algorithm, we need a dataset. So during the first step of KNN, we must load the training data as well as the test data.

Step 2 Next, we need to choose the value of K, i.e., the nearest data points. K can be any integer.

Step 3 For each point in the test data, do the following:

- Calculate the distance between the test data and each row of training data with the help of any method, namely: Euclidean, Manhattan, or Hamming distance. The most common method for calculating distance is the Euclidean method.
- Based on the distance value, sort them in ascending order.
- Next, it will choose the top K rows from the sorted array.
- Now, it will assign a class to the test point based on the most frequent class in these rows.

The Third Algorithm:

The Decision Tree Classifier algorithm was used. In a decision tree, to predict the class of the given dataset, the algorithm starts from the root node of the tree. This algorithm compares the values of the root attribute with the attribute of the record (the real dataset), and based on the comparison, it follows the branch and moves to the next node.

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The Fourth Algorithm:

The Random Forest Classifier algorithm was used. It is a supervised learning algorithm. The "forest" that it builds is a set of decision trees, usually trained with the "bagging" method. The general idea of the bagging method is that a mixture of learning models increases the overall result. One of the great advantages of Random Forest is that it can be used in classification and regression problems, which make up the majority of current machine learning systems.

3-1-7 Comparing the Results and Choosing the Best Algorithm

In this stage we compare the results we obtained from each algorithm and choose the algorithm with the best values.

- The table shows the results of the first algorithm, SVM:

Metric	Value
Accuracy Of Training	100%
Accuracy Of Testing	50%
MSE (Mean Squared Error)	1.282548
MAE (Mean Absolute Error)	0.756232

We note the decrease in test accuracy, and the reason is due to the way the algorithm works, as it is considered linear and it is weak in solving real-world problems.

- The table shows the results of the second algorithm, KNeighbors Classifier:

Metric	Value
Accuracy Of Training	82%
Accuracy Of Testing	88%
MSE (Mean Squared Error)	0.581717
MAE (Mean Absolute Error)	0.307479

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These are acceptable values, but other methods were tried to make sure of obtaining the best result.

- The table shows the results of the third algorithm, Decision Tree Classifier:

Metric	Value
Accuracy Of Training	99.3%
Accuracy Of Testing	83%
MSE (Mean Squared Error)	0.530470
MAE (Mean Absolute Error)	0.289473

These are acceptable values, but other methods were tried to make sure of obtaining the best result.

- The table shows the results of the fourth algorithm, Random Forest Classifier:

Metric	Value
Accuracy Of Training	100%
Accuracy Of Testing	86.8%
MSE (Mean Squared Error)	0.48476
MAE (Mean Absolute Error)	0.24930

And through the comparison we note that all the algorithms, except the first algorithm, gave accuracy results higher than 80%, and this is because of their nonlinearity.

The Random Forest Classifier algorithm was adopted, as it has the best results with 86% for accuracy and 0.249 for error.

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3-2 Discussion and Comparison of the Results:

Our proposed program was compared with a study conducted in 2019 by Sriwijaya University in Indonesia. As shown in the table, our proposed study was compared with the Sriwijaya study in terms of the database used, the classes used, the method followed, the hardware resources used, and the accuracy reached.

[Figure: bar chart of training and test accuracy (0–120) for SVM, KNeighbors Classifier, Decision Tree Classifier, Random Forest Classifier. Legend: Training; Test]

[Figure: bar chart of MSE and MAE (0–1.4) for SVM, KNeighbors Classifier, Decision Tree Classifier, Random Forest Classifier. Legend: MSE; MAE]

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	Our proposed study	Sriwijaya University study
Database	MIT-BIH	MIT-BIH
Classes used	The algorithm classifies 3 types: PVC-LBBB-NORMAL	The algorithm classifies 9 types: APC-LBBB-N-PACED-RBBB PVC-FV-VFW-NE
Method followed	An algorithm was designed to extract the features of ECG signals and process them using ML algorithms	The signals were processed and then DAEs networks were used for unsupervised learning, and also a DNN network as a classifier
Hardware resources used	Core i5-7200U, 2 GB graphics card, 8 GB RAM (memory), and 1000 GB HDD	ASUS TH server, 8th Generation, 32 GB RAM, 500 GB SSD hard drive, and NVIDIA 12 GB graphics card
Accuracy reached	An accuracy of 86.7% was reached	Accuracy of 91.73%

Advantages:

1- The built model gives freedom in choosing the important features for each case after studying it physiologically.

2- High accuracy after processing the manually extracted features, compared with extraction in the case of neural networks.

Drawbacks:

The small amount of data used in training and testing, and thus, despite the high accuracy, the model cannot be generalized on a wide scale, because it is likely that the accuracy is not sufficient for dealing with larger and more general data.

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Future Aspirations and Proposals:

1- Expanding the database to include more than 3 cases of the cardiac signal.

2- A convolutional neural network can be integrated with our program in order to give higher accuracy and greater flexibility.

3- Giving the program greater flexibility that allows it to read the cardiac signal from more than one lead, as it is now programmed to read it from a single lead, which is Lead II.

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Note: This is a translated copy only.